MINIMAL NON-WEIERSTRASS SEMIGROUPS

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ABSTRACT. Let $p \in X$ be a point on a compact Riemann surface. The *Weierstrass* semigroup of p is the semigroup of pole orders of meromorphic functions on X that are regular at all but p. Hurwitz asked in 1892 whether all numerical semigroups occur as Weierstrass semigroups.

In this paper we give a new method for showing that certain numerical semigroups are not Weierstrass, including some of every genus g in which non-Weierstrass examples could exist, except $g = 18$. Our example for $g = 13$ has at once the smallest possible genus, multiplicity and number of generators of any possible non-Weierstrass semigroup.

Our technique applies to prove that certain semigroups are not Weierstrass semigroups over any field.

INTRODUCTION

A *numerical semigroup* is a subset of the non-negative integers with finite complement that contains 0 and is closed under addition. To simplify the language we generally drop the word "Numerical".

If S is a semigroup then the *gaps* of S are the positive integers not in S and the cardinality g of the set of gaps is called the *genus* of the semigroup. The smallest nonzero element m of S is the *multiplicity* and the minimal number of generators is the *embedding dimension* of the semigroup algebra $k[t^s \mid s \in S]$, where k is a field. The semigroup is called Weierstrass if it occurs as the semigroup of pole orders of rational functions that are regular except at a point p of a compact Riemann surface X; by Weierstrass' Lückensatz (an easy consequence of the Riemann-Roch theorem) the genus of X coincides with the genus of the semigroup.

The main contribution of this paper is a condition on the free resolution of a semigroup ideal that forces every positively graded deformation to have a certain multivalued section. Under an additional hypothesis, this multisection tracks singular points of each fiber, proving that no deformation can be smooth. By Pinkham's Theorem [Pin74, Chapter IV], this implies that the semigroup is not Weierstrass. We call a semigroup satisfying these conditions *special* (Definition 2.2).

On the other hand, we show that every semigroup of genus < 13 is Weierstrass. There are 1413 such semigroups [ESa], or https://oeis.org/A007323. Results in the literature, quoted below, prove that 781 of them

occur as Weierstrass semigroups. In an auxiliary file (see below) we give explicit positively graded smoothing families for the other 632, found using computational methods that we explain in Section 5. By Pinkham's Theorem [Pin74], this completes the proof that all 1413 examples occur as Weierstrass semigroups.

Since every semigroup of genus < 13 is Weierstrass, as is every semigroup of multiplicity < 6 or embedding dimension < 4 (see the history below), our genus 13 example, which is generated by

$$\{6, 9, 13, 16\}$$

has the smallest possible genus, multiplicity, and embedding dimension of any non-Weierstrass semigroup. In Theorem 3.1, we exhibit 4-generator semigroups of multiplicity 6 and every genus $g \geq 13$ except $g = 15$ and $g = 18$. For $g = 15$ we give an example of multiplicity 8.

We have put the Macaulay2 code necessary to construct and verify our assertions in the GitHub repository **eisenbud/WeierstrassSemigroups** https://github.com/eisenbud/WeierstrassSemigroups. In the file **smoothingFamilies.dbm** we give smoothing families for the 632 semigroups that were not previously known to be Weierstrass. To check the flatness and smoothness of these families, simply execute the code in the file **allSemigroupsOfGenusLessThan13AreWeierstrass.m2**. The code that we have used, both to test whether a semigroup is degree-special and to prove that certain families we have constructed are special, is in **NumericalSemigroups.m2** [ESa] and **WeierstrassSemigroups.m2** [ESb]. These packages are distributed with the Macaulay2 system [GS].

Background and history. Suppose that X is a compact Riemann surface, or more generally a smooth projective curve over an algebraically closed field, of genus $g \geq 2$. Since there are no everywhere regular analytic functions on X except constants, the nonconstant meromorphic functions that are regular except at a point $p \in X$, must have poles at p. Since the pole order of a product of two such functions is the sum of their pole orders, and there are always functions regular on $X \smallsetminus \{p\}$ with pole order exactly $2g + c$ for any $c \geq 0$, the pole orders form a *numerical semigroup* $S := S(X, p)$.

In his lectures in the 1860's Weierstrass proved that, for every $p \in X$, the semigroup $S(X, p)$ contains all but exactly $g :=$ genus X positive integers, and is thus a numerical semigroup of genus g. The result was published by Weierstrass' student Schottky in [Sch77]. Following Haure [Hau96] the semigroup $S(X, p)$ is now called the Weierstrass Semigroup of X at p.

Hurwitz [Hur92] used Weierstrass points to prove that the automorphism group of a Riemann surface is finite, and posed the problem of deciding which semigroups actually occur as Weierstrass semigroups. The problem

attracted immediate attention: Haure gave an incorrect criterion for a semigroup to occur, excluding for example the Weierstrass semigroup $\{4, 6, 19\}$ [Hau96, p, 165]. Hensel-Landsberg [HL02, pp. 545–548] took up the problem, and gave an approach to constructing a Riemann surface with any given semigroup using singular plane curves with properties that cannot always be achieved. (We are grateful to Jan Stevens for clarifying this for us.)

There, it seems, the matter rested until Henry Pinkham [Pin74] gave an algorithm using deformation theory that can in principle determine whether a given semigroup is Weierstrass (see Section 1).

Using deformations of curves of compact type, Eisenbud and Harris [EH87] proved that in characteristic 0 all semigroups of low weight are Weierstrass. That result was extended to all characteristics by Osserman [Oss06] and to a wider class of semigroups by Pflueger [Pfl18]. It was known that every semigroup having genus < 8 ([Kom94]) or multiplicity < 6 [Kom92] or number of generators < 4, and every symmetric semigroup with 4 generators, is Weierstrass [Wal82].

On the other hand, in [Buc81] in 1980, Buchweitz gave the first example of a numerical semigroup that is not Weierstrass. It is generated by the elements

$$\{13, 14, 15, 16, 17, 18, 20, 22, 23\}$$

and has genus 16. (Buchweitz's proof: the number of integers that can be expressed as the sum of a pair of gaps of S is 46, while the dimension of the space of quadratic differential forms on a surface of genus 16 is only 45.) Since 1980 many other semigroups have been shown not to be Weierstrass, both using Buchweitz's method and otherwise; see for example [GGMATVT21], [Kom21], and [Kom13]. Examples by Komeda [Kom13] had, until now, the smallest known multiplicities, 8 and 12. Until this paper, no non-Weierstrass semigroup of genus < 16 or multiplicity < 8 was known.

For more on the historical impact of the theory of Weierstrass points, see [DC08]

Notation. We generally define a semigroup S by giving an ordered set of generators $\{s_0 < \cdots < s_u\}$. Abusing notation, we then write $S = \{s_0, \ldots, s_u\}$. If S is a semigroup, and k is a field, then the *semigroup algebra* of S over k is the graded ring $k[S] := k[\{t^s \mid s \in S\}] \subset k[t]$. The *semigroup ideal* is the kernel of the natural map from the polynomial ring on variables corresponding to the t^{s_i} to $k[S]$.

When we write a semigroup ring as a quotient of a polynomial ring, we use the $\mathbb{Z}$-grading induced by the semigroup. We order the variables by increasing degree, and label them by the residue class of their degree modulo the multiplicity. Thus for example the semigroup ring corresponding to the

semigroup generated by $\{6, 9, 13, 16\}$ over a field k is written as a quotient of the polynomial ring $k[x_0, x_3, x_1, x_4]$.

If M is a map between free modules, or a matrix representing such a map, we write $I_j(M)$ for the ideal generated by the $j \times j$ minors (= determinants of $j \times j$ submatrices) of M.

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1. PINKHAM'S THEOREM AND A STRUCTURE THEOREM

In this section we review two known results that we will need.

1.1. **Pinkham's Theorem.**

Pinkham [Pin74, Theorem 13.9] gives a criterion for a semigroup to be Weierstrass:

Proposition 1.1 (Pinkham)**.** *Let S be a numerical semigroup of genus g. S is a Weierstrass semigroup if and only if there exists a one-parameter positively graded smoothing of the semigroup ring $k[S]$ of S.*

By *a positively graded smoothing* we mean a positively graded algebra whose grading extends the grading induced by the semigroup, and whose general fiber is a smooth affine curve. This is what Pinkham would call a smoothing with negative grading (Pinkham's sign comes from the natural grading on the vector space T^1 of first-order infinitesimal deformations). For the reader's convenience we give a proof:

Proof. Suppose S is the Weierstrass semigroup of a point p on a smooth projective curve C. Consider the line bundle $\mathcal{O}_C(p)$ and the graded ring

$$A = \bigoplus_{i \geq 0} H^0(C, \mathcal{O}_C(ip)).$$

As a k-algebra, A is generated by elements of degree s_i corresponding to generators of the semigroup S and the element $z \in H^0(C, \mathcal{O}_C(p))$ corresponding to the constant function 1. Choosing appropriate generators we see that $A/z \cong k[S]$. Since $C = \operatorname{Proj} A$ is smooth the open subset, where $z \neq 0$ is smooth, and since z is a non-zero divisor A is a flat $k[z]$-algebra.

Conversely, if

$$\begin{array}{ccc} \operatorname{Spec} k[S] & \longrightarrow & \operatorname{Spec} A \\ \downarrow & & \downarrow \\ \operatorname{Spec} k & \longrightarrow & \operatorname{Spec} k[z] \end{array}$$

is a homogeneous one-parameter smoothing, then after base change we may assume that $\deg z = 1$. Then $C = \operatorname{Proj} A$ is a smooth projective curve in a weighted projective space with a unique point p at infinity, which has S as Weierstrass semigroup. Note that C is smooth because the general point of the special fiber of the affine curve $\operatorname{Spec} k[S]$ is smooth; the singular point corresponds to the irrelevant maximal ideal of A. □

All the flat, positively graded families of $k[z]$-algebras specializing to $k[S]$ are flat subfamilies of the universal positively graded unfolding of $k[S]$. If there is a natural multisection defined over the unfolding whose values are singular points, Pinkham's theorem shows that S is not Weierstrass. Conversely if there are singular points in every fiber then they define a multisection. Thus in proving that a semigroup is not Weierstrass, it is natural to look for a multisection. However there are also families with section where the section actually follows the "origin", tracing out a smooth point—see Example 3.2.

1.2. **A structure theorem for resolutions.** Here is a case of [BE74, Theorem 3.1] and [Bru87], that we will use. It is closely related to the "determinant of a complex" discovered by Cayley [Cay48] and also exploited in [KM76] and [Mac65].

Proposition 1.2. *Suppose that* $F_0 \xleftarrow{\psi_1} F_1 \xleftarrow{\psi_2} F_2 \xleftarrow{\psi_3} F_3 \longleftarrow 0$ *is the free resolution of a module of grade* ≥ 2 *over a local or positively graded ring* R. *Writing* $r_i = \operatorname{rank}(\psi_i)$, *there is a commutative diagram*

$$\begin{array}{ccccc} \bigwedge^{r_1} F_1 \cong \bigwedge^{r_2} F_1^* & & \xrightarrow{\quad\bigwedge^{r_2}\psi_2^*\quad} & & \bigwedge^{r_2} F_2^* \cong \bigwedge^{r_3} F_2 \ . \\ & \searrow^{\bigwedge^{r_1}\psi_1} & & \nearrow_{\bigwedge^{r_3}\psi_3} & \\ & & \bigwedge^{r_1} F_0 \cong R \cong \bigwedge^{r_3} F_3 & & \end{array}$$

In particular

$$I_{r_2}(\psi_2) = I_{r_1}(\psi_1) I_{r_3}(\psi_3)$$

2. Special resolutions and Non-Weierstrass Semigroups

The *format* of a free complex

$$F_0 \longleftarrow F_1 \longleftarrow \cdots$$

is the list of integers $\{\text{rank } F_0, \text{rank } F_1, \cdots\}$ where we suppress the ranks corresponding to any terms $F_i = 0$ in the complex.

Definition 2.1. We will say that a minimal resolution

$$*) \qquad P \xleftarrow[\phi_1]{} P^6 \xleftarrow[\phi_2]{} P^8 \xleftarrow[\phi_3]{} P^3 \longleftarrow 0$$

over a ring P is *special of format* $\{1, 6, 8, 3\}$ if it admits an acyclic subcomplex

$$**) \qquad P \xleftarrow[\phi_1']{} P^4 \xleftarrow[\phi_2']{} P^4 \xleftarrow[\phi_3']{} P \longleftarrow 0.$$

that is, term by term, a summand of $*)$.

In matrix terms, this means that there is a choice of generators of the free modules of the complex $*)$ such that:

(1) The last 4 entries in the first column of the matrix ϕ_3, are zero as in the diagram below, where we show the column as the first column:

$$\phi_3 : \quad \begin{pmatrix} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ 0 & * & * \\ 0 & * & * \\ 0 & * & * \\ 0 & * & * \end{pmatrix},$$

and ϕ_3' is the 4×1 submatrix corresponding to the first 4 elements in this column.

(2) We may divide ϕ_2 into two 6×4 matrices, $\phi_2 = A \mid B$, such that the 5th and 6th rows of A are zero as in the following diagram:

$$A : \quad \begin{pmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

and ϕ_2' is the 4×4 matrix corresponding to the first 4 rows of A.

(3) With these choices, ϕ_1' is the matrix corresponding to the first 4 generators of I

To show that some semigroups with special resolution of format $\{1,6,8,3\}$ are not Weierstrass, we need conditions that ensure that the $\{1,4,4,1\}$ subcomplex persists in flat graded deformations.

Definition 2.2. A numerical semigroup S is *special* if every flat positively graded deformation of the semigroup ring $k[t^S]$ has minimal free resolution that is special of format $\{1,6,8,3\}$.

Proposition 2.3. *With notation as above, if *) is special of format* $\{1,6,8,3\}$ *and the image of* ϕ_1 *is a prime ideal then, with notation as above, the four entries of the matrix* ϕ_3' *are a regular sequence, and thus define a codimension 4 subscheme of* $\operatorname{Spec} P$*. Thus if the semigroup* S *is special, then the entries of* ϕ_3' *define a finite multisection in every deformation of* $\operatorname{Spec} k[S]$.

Proof. Let J be the ideal of P generated by the 4 entries of ϕ_3'. The cokernel of the dual of ϕ_3 is, up to a shift in grading, the canonical module ω of $k[S]$, and thus its annihilator in P is the annihilator I of $k[S]$, which is prime. The ideal J is the ideal quotient of one of the generators of ω into the other two, and thus $I \subset J$. Since I requires 6 generators, $I \neq J$, and thus J has codimension 4. Since the resolution *) is minimal, the 4 generators of J are of positive degree, and it follows that they form a regular sequence, proving the first statement. The second statement follows at once. □

The definition of special requires information about every flat positively graded deformation of the semigroup ring, but we can give a sufficient condition on the minimal graded resolution of the semigroup ring alone:

In the matrix representing a homogeneous map of free modules $\oplus P(c_j) \to \oplus P(r_i)$ we say that the element in position (i,j) has *formal degree* $c_j - r_i$. If the element is nonzero, this is the ordinary degree. In our case P will be positively graded, so if an element of a matrix has negative formal degree, then the element is zero.

Proposition 2.4. *Let* S *be a numerical semigroup, and write the semigroup ring* $k[t^S]$ *of* S *as* P/I*, where* P *is a graded polynomial ring whose generators have degrees corresponding to the generators of* S*. If the minimal graded* P*-free resolution of the semigroup ring* $k[t^S] = P/I$ *has the form*

$$P \xleftarrow[\phi_1]{} P^6 \xleftarrow[\phi_2]{} P^8 \xleftarrow[\phi_3]{} P^3 \longleftarrow 0,$$

and (possibly after reordering the generators of the free modules) satisfies the following conditions, then S *is special.*

1. *The last four entries of the first column of the matrix* ϕ_3 *have negative formal degree.*
2. *The first three entries of each of the last two rows of* ϕ_2 *have formal degree strictly less than the multiplicity of* S*.*

(3) *The induced subcomplex*

$$P \xleftarrow[\phi_1']{} P^4 \xleftarrow[\phi_2']{} P^4 \xleftarrow[\phi_3']{} P \longleftarrow 0,$$

where ϕ_1' consists of the first 4 columns of ϕ_1, ϕ_2' consists of the first 4 rows and columns of ϕ_2 and ϕ_3' consists of the first four rows of the first column of ϕ_3, is acyclic.

Moreover, when these conditions are met, the 4 elements of the matrix ϕ_3' form a regular sequence.

Proof of Proposition 2.4. The elements of negative formal degree are of course 0.

Write $\omega := \operatorname{coker}\phi_3^*$. Since ω is, up to a shift in grading, the canonical module of P/I, the annihilator of ω is I. Let w_i be the generator of ω corresponding to the i-th column of ϕ_3. The ideal J generated by the first column of ϕ_3 is the annihilator of the image of w_1 in $\omega/(Pw_2 + Pw_3)$, so $I \subset J$. By the special hypothesis, J is generated by 4 elements, whereas I requires 6 generators, so $I \subsetneq J$. As I is a prime of codimension 3 J must have codimension 4, so the 4 nonzero elements in the first column of ϕ_3 are a regular sequence, as claimed.

The degrees of these four elements are in the semigroup S, and since they form a regular sequence, any nonzero relation among them must have degrees in S too. The first 4 elements in each row of ϕ_2 form such a relation. By the special hypothesis, the first 3 elements of the last two rows of ϕ_2 are zero, and since they are relations on a regular sequence the 4th element in each of these rows must be zero as well. □

Definition 2.5. We call a semigroup with format $\{1, 6, 8, 3\}$ *degree-special* if it satisfies the conditions (1), (2) and (3) of Proposition 2.4; this implies that the semigroup is special.

The following conjecture is based on our examination of hundreds of examples:

Conjecture 2.6. *If S satisfies conditions (1) and (2), then it satisfies condition (3) as well; so S is special without further hypotheses.*

Example 2.7. The semigroup S generated by $\{6, 9, 13, 16\}$ is degree-special. Indeed, the semigroup ring of S has differentials

$$\phi_1 = \begin{pmatrix} x_0^3 - x_3^2 & x_3x_1 - x_0x_4 & x_0^2x_1 - x_3x_4 & x_0x_1^2 - x_4^2 & x_0^2x_3^3 - x_1^3 & x_0x_3^4 - x_1^2x_4 \end{pmatrix}$$

$$\phi_2 = \begin{pmatrix} -x_1 & -x_4 & 0 & 0 & 0 & -x_0x_3^3 & 0 & x_0x_3^2x_4 \\ -x_3 & -x_0^2 & -x_4 & -x_0x_1 & -x_1^2 & 0 & -x_0x_3^3 & x_0^3x_3^2 \\ x_0 & x_3 & -x_1 & -x_4 & 0 & x_1^2 & 0 & 0 \\ 0 & 0 & x_0 & x_3 & 0 & 0 & 0 & -x_1^2 \\ 0 & 0 & 0 & 0 & -x_3 & x_0^2 & -x_4 & -x_0x_1 \\ 0 & 0 & 0 & 0 & x_0 & -x_3 & x_1 & x_4 \end{pmatrix}$$

$$\phi_3 = \begin{pmatrix} x_4 & x_0x_3^3 & 0 \\ -x_1 & -x_0^2x_3^2 & 0 \\ -x_3 & -x_1^2 & 0 \\ x_0 & 0 & -x_1^2 \\ 0 & x_4 & x_0x_1 \\ 0 & -x_1 & -x_4 \\ 0 & -x_3 & -x_0^2 \\ 0 & -x_0 & -x_3 \end{pmatrix}$$

and satisfies the degree conditions.

Proposition 2.8. *Suppose that P is a local or positively graded ring, and that*

$$P \xleftarrow[\psi_1]{} P^4 \xleftarrow[\psi_2]{} P^4 \xleftarrow[\psi_3]{} P \xleftarrow{} 0$$

is a minimal free resolution of format $\{1,4,4,1\}$. If the 4 entries of the matrix ψ_3 form a regular sequence, then the entries of the matrix ψ_2 are in the ideal J generated by the entries of ψ_3, and the entries of ψ_1 are in J^2.

Proof. Since $I := I_1(\psi_1)$ is an ideal of finite homological dimension, [BE74, Corollary 5.2] shows that we may write $I = rI'$ for some nonzerodivisor r and some ideal I' of depth ≥ 2. Thus, replacing ψ_1 by $\psi_1' := r^{-1}\psi_1$ if necessary, we may assume that $*)$ is a resolution of a module of grade ≥ 2.

The first assertion of the Proposition holds because the columns of ψ_2^* are syzygies of the regular sequence of elements of ψ_3^*, and are thus linear combinations of the Koszul syzygies.

By Proposition 1.2 we have $I_3(\psi_2) = I_1(\psi_1')J$. By the first assertion, $I_3(\psi_2) \subset J^3$. Thus

$$I \subset I_1(\psi_1') \subset I_3(\psi_2) : J \subset J^3 : J.$$

Since J is a complete intersection,

$$J^3 : J = J^2,$$

by [Eis95, Exercise 17.13 e)] completing the proof. (The last equality also follows directly from the well-known fact that the Rees algebra of the ideal generated by a regular sequence is isomorphic to a polynomial ring.) □

Theorem 2.9. *If S is a special semigroup then S is not a Weierstrass semigroup.*

Proof. We will show that any graded deformation of $\operatorname{Spec} k[t^S]$ is singular along the distinguished multisection of Definition 2.2. Proposition 1.1 shows that S is not Weierstrass.

With notation as in the definition of special, we consider the minimal free resolution of the semigroup ring P/I. The condition that the last four entries of the first column of ϕ_3 have negative formal degree persists in any flat deformation. By Proposition 2.4 the first four entries of that column form a regular sequence, and this condition too persists. Also the degree condition on the first four entries of the last two rows of ϕ_2 persists, and implies that these entries are zero. Thus there is a $\{1,4,4,1\}$ subcomplex in any deformation. By condition (3) of Proposition 2.4, the subcomplex of format $\{1,4,4,1\}$ in the resolution of the semigroup ring is acyclic. Since acyclicity is an open condition, this persists in every deformation.

To complete the proof we will show, more generally, that if P is a positively graded polynomial ring, and P/I is a factor ring with minimal free resolution of special format $\{1,6,8,3\}$ then with ideals $I_1(\psi_1), I_3(\psi_2), J = I_1(\psi_3)$ defined in terms of the distinguished $\{1,4,4,1\}$ subcomplex as in the proof of Proposition 2.8, $\operatorname{Spec} P/I$ is singular along $\operatorname{Spec} P/J$.

This follows because $I_1(\psi_1)$ is generated by four of the 6 generators of I and lies in J^2. Thus all partial derivatives of the polynomials generating $I_1(\psi_1)$ are in J, so the 3×3 minors of the Jacobian matrix of I also lie in J proving the claim. □

Example 2.10. Returning to the example of the semigroup generated by $\{6,9,13,16\}$ above we see that the induced subcomplex of format $\{1,4,4,1\}$ has differentials

$$\phi_1' = \begin{pmatrix} x_0^3 - x_3^2 & x_3x_1 - x_0x_4 & x_0^2x_1 - x_3x_4 & x_0x_1^2 - x_4^2 \end{pmatrix}$$

$$\phi_2' = \begin{pmatrix} -x_1 & -x_4 & 0 & 0 \\ -x_3 & -x_0^2 - x_4 & -x_0x_1 & -x_1^2 \\ x_0 & x_3 & -x_1 & -x_4 \\ 0 & 0 & x_0 & x_3 \end{pmatrix}, \quad \phi_3' = \begin{pmatrix} x_4 \\ -x_1 \\ -x_3 \\ x_0 \end{pmatrix}.$$

Using the theorem of [BE73] it is easy to show that this subcomplex is acyclic. In particular S is not a Weierstrass semigroup.

There are several ways to check condition (3), the acyclicity of the $\{1,4,4,1\}$ subcomplex, in the definition of degree-special:

Proposition 2.11. *Suppose that S is a numerical semigroup whose minimal free resolution satisfies conditions (1) and (2) in Definition 2.5, and let*

$$P \xleftarrow[\phi_1']{} P^4 \xleftarrow[\phi_2']{} P^4 \xleftarrow[\phi_3']{} P \longleftarrow 0$$

be the subcomplex of the minimal free resolution of $P/I = k[t^S]$ as in the proof of Proposition 2.4. Each of the following hypotheses implies condition (3) of Proposition 2.4.

(1) *The 3×3 minors of ϕ_2' generate an ideal of codimension ≥ 2.*
(2) *The 2×2 minors of the 4×2 submatrix of ϕ_3 consisting of the last 4 rows of the last 2 columns generate an ideal of codimension ≥ 2.*
(3) *After possibly reordering the rows and columns of ϕ_2', the ϕ_i' satisfy the following 3 conditions, (compare Example 2.10):*
   (a) *The first two elements of the last row of ϕ_2' are zero.*
   (b) *the last two elements of the last row of ϕ_2' form a regular sequence.*
   (c) *the 2×2 minors of the first two columns of ϕ_2' are nonzero and have formal degrees equal to the formal degrees of the first 3 generators of the semigroup ideal.*

Proof. The sufficiency of Hypothesis (1) follows from the main theorem of [BE73].

Hypothesis (2) $\Rightarrow$ Hypothesis (1): As shown in the diagram below, we divide the three matrices ϕ_1, ϕ_2, ϕ_3 into blocks. In the diagram below A is a 1×4 block, so A^t, the transpose, is a 4×1 block. Similarly, $B = \phi_2'$ has size 4×4, $\widetilde{B}$ has size 2×4, C^t has size 1×4 and $\widetilde{C}^t$ has size 2×4. The remaining blocks have entries 0 or entries $*$ that do not play a role in the argument below.

$$\begin{array}{|c|c|}\hline \phi_1^t & \phi_2 \\ \hline & \phi_3^t \\ \hline \end{array} = \begin{array}{|c||cccc|cccc|}\hline A^t & & B & & & * & * & * & * \\ & & & & & * & * & * & * \\ & & & & & * & * & * & * \\ & & & & & * & * & * & * \\ \hline * & 0 & 0 & 0 & 0 & & & & \\ * & 0 & 0 & 0 & 0 & & \widetilde{B} & & \\ \hline & & & C^t & & 0 & 0 & 0 & 0 \\ \hline & * & * & * & * & & & & \\ & * & * & * & * & & \widetilde{C}^t & & \\ \hline \end{array}$$

By Proposition 1.2 we have

$$I_3(B)I_2(\widetilde{B}) = I_1(A)I_1(C)I_2(\widetilde{C}).$$

Under Hypothesis (2), $I_2(\widetilde{C})$ has depth ≥ 2. Since I is prime, every minimal generator of I is prime, and thus $I_1(A)$ has depth ≥ 2. Moreover, $I_1(C)$ has

depth 4, and thus $I_3(B) \supset I_3(B)I_2(\widetilde{B})$ has depth ≥ 2 completing the proof in this case.

Hypothesis (3) $\Rightarrow$ Hypothesis (1): We now subdivide the 4×4 matrix $B = \phi_2'$ into 4 blocks

$$B = \begin{array}{|cc|cc|} \multicolumn{2}{|c|}{\multirow{3}{*}{B_1}} & * & * \\ & & * & * \\ & & * & * \\ \hline 0 & 0 & \multicolumn{2}{|c|}{B_2} \\ \hline \end{array}$$

We see that $I_3(B) \supseteq I_2(B_1)I_1(B_2)$. Since we have assumed that $I_1(B_2)$ has depth 2, it suffices to prove the same for $I_2(B_1)$

Let B_1 be the 3×2 submatrix of B consisting of the first two columns and the first 3 rows. By assumption the 2×2 minors of B_1 are not all zero, and thus generate an ideal of depth ≥ 1. Since each minimal generator of I is prime, the first three generators of I define an ideal of depth ≥ 2. It follows from the main theorem of [BE73] that the image of the column matrix made of these three generators is the kernel of the transpose of B_1^t.

On the other hand, Cramer's rule implies that the column matrix of the 2×2 minors of B_1 composes with B_1 to 0, and thus it is a multiple of the column of the first three generators of I. It follows that the 2×2 minors of A are multiples of these generators, and since they have the same degrees, they must generate the same ideal, completing the proof. □

3. Degree-special semigroups of larger genus

The following result covers all genera ≥ 13 except $g = 18$. Direct enumeration shows that no degree-special semigroup has genus 18 (we have no conceptual understanding of this gap).

Theorem 3.1. *For $g \geq 13$ but $g \not\equiv 0 (\mathrm{mod}\ 3)$, the semigroup $\{6, 9, g, g + 3\}$ is degree-special of genus g. For $g \geq 21$ and $g \equiv 0 (\mathrm{mod}\ 3)$, the semigroup $\{6, 15, g - 1, g + 2\}$ is degree-special of genus g. The semigroup $\{8, 10, 13, 15\}$ is degree-special of genus 15.*

Proof. We give the details for the case of the semigroups $\{6, 9, g, g+3\}$ with $g \equiv 2$ or $4 (\mathrm{mod}\ 6)$, the other cases being similar.

The semigroup ideal of L has a $\mathbb{Q}[x_0, x_3, y, w]$-resolution

$$0 \longleftarrow S \xleftarrow{\phi_1} S^6 \xleftarrow{\phi_2} S^8 \xleftarrow{\phi_3} S^3 \longleftarrow 0\ .$$

In case of $g \equiv 2$ or $4 (\mathrm{mod}\ 6)$ the differentials are as indicated in Table 1.

Here we could use any monomial order on S for which the terms colored in red in the first matrix are the lead terms (for example we could take the weight order of the grading on S refined by the reversed lexicographic order with $x_3 > y > w > x_0$). The red terms in the syzygy matrices are then the

$$\begin{array}{|c|c|}\hline \phi_1^t & \phi_2 \\ \hline & \phi_3^t \\ \hline \end{array} = \begin{array}{|c|cccccccc|}\hline x_3^2-x_0^3 & -y & -w & 0 & 0 & 0 & f & 0 & 0 \\ x_3y-w\,x_0 & x_3 & x_0^2 & -w & y\,x_0 & -y^2 & 0 & -f & 0 \\ x_3w-y\,x_0^2 & x_0 & x_3 & y & -w & 0 & -y^2 & 0 & f \\ w^2-y^2x_0 & 0 & 0 & -x_0 & x_3 & 0 & 0 & 0 & -y^2 \\ y^3-x_0f & 0 & 0 & 0 & 0 & x_3 & -x_0^2 & w & -y\,x_0 \\ y^2w-x_3f & 0 & 0 & 0 & 0 & -x_0 & x_3 & -y & w \\ \hline & w & -y & x_3 & x_0 & 0 & 0 & 0 & 0 \\ & f & 0 & y^2 & 0 & -w & y & x_3 & -x_0 \\ & 0 & -f & 0 & y^2 & yx_0 & -w & x_0^2 & x_3 \\ \hline \end{array}$$

where $f \in k[x_0]$.

TABLE 1. Syzygy matrices for semigroups $\{6, 9, g, g+3\}$ for $g \equiv 2$ or $4 \pmod 6$.

lead term with respect to the induced monomial orders, and the complex is exact by the Gröbner basis algorithm for syzygies as presented for example in [Sch25][Section 8.3]. In this case the exactness of the $\{1, 4, 4, 1\}$ subcomplex follows as well. In the universal family the entries indicated in blue remain zero for degree reasons if $g \geq 13$. □

Table 2 gives the complete list of the 4-tuples of generators of degree-special semigroups up to genus 20:

Genus	Generators
13	$\{6,9,13,16\}$, $\{8,9,12,13\}$
14	$\{6,9,14,17\}$
15	$\{8,10,13,15\}$
16	$\{6,9,16,19\}$, $\{8,11,12,15\}$
17	$\{6,9,17,20\}$, $\{9,10,14,15\}$
19	$\{6,9,19,22\}$, $\{8,12,13,17\}$
20	$\{6,9,20,23\}$, $\{6,15,19,22\}$

TABLE 2. All the degree-special semigroups of genus $g \leq 20$.

Example 3.2. As explained in the introduction, we could search for semigroups that are not Weierstrass by looking for those moving in families with a multisection. However, there are positively graded deformations of semigroup rings that admit a section, but where the image of the section consists of smooth points of the fibers. This is the case for the semigroup $L = \{7, 15, 23, 39\}$. The genus of the semigroup is 30. The minimal free

resolution over $S = \mathbb{Q}[x_0, x_1, x_2, x_4]$ has shape

$$0 \longleftarrow S \xleftarrow{\phi_1} S^4 \xleftarrow{\phi_2} S^6 \xleftarrow{\phi_3} S^3 \longleftarrow 0$$

with differentials as indicated below.

ϕ_1^t	ϕ_2
	ϕ_3^t

$=$

$x_1^2 - x_0x_2$	$-x_2^2 + x_0x_4$	$-x_2x_4$	$-x_0^{10}$	$-x_4^2 + x_0^9x_1$	$x_0^9x_2$	0
$x_2^2 - x_0x_4$	$x_1^2 - x_0x_2$	$-x_0x_4$	$-x_1x_4$	0	x_0^{10}	$x_0^9x_1 - x_4^2$
$x_1x_2x_4 - x_0^{11}$	0	x_1	x_2	0	$-x_4$	0
$x_4^2 - x_0^9x_1$	0	$-x_0^2$	$-x_0x_1$	$x_1^2 - x_0x_2$	x_1x_2	$x_2^2 - x_0x_4$
	x_4	$-x_2$	x_1	x_0	0	0
	0	x_4	0	$-x_2$	x_1	$-x_0$
	$-x_0^9$	0	$-x_4$	0	$-x_2$	x_1

Again the lead terms in suitable monomial orders are indicated in red. The universal unfolding has a section because the entries indicated by blue remain zero for degree reasons. However this is a Weierstrass semigroup by [Wal80], since the semigroup ring is an almost complete intersection. Indeed a flat smoothing family is given by the polynomials

$$\begin{aligned}
&x_1^2 - x_2x_0 + x_2z^7\\
&x_2^2 - x_4x_0\\
&x_1x_2x_4 - x_0^{11} + x_0^{10}z^7 - x_0^2z^{63} + x_0z^{70}\\
&x_4^2 - x_1x_0^9 - x_1z^{63}.
\end{aligned}$$

In this family, the value of the section is always the point defined by the ideal (x_0, x_1, x_2, x_4). Computational steps to produce such families are described in Section 5.

4. Further examples

Example 4.1. Let $L = \{6, 9, 13, 16\}$ and consider the semigroup generated by $L' = 2L \cup \{6 + 9\} = \{12, 15, 18, 26, 32\}$. The semigroup ideal $I_{L'} \subset S = \mathbb{Q}[x_0, x_3, x_6, x_2, x_8]$ is generated by the image of $I_L \subset \mathbb{Q}[x_0, x_3, x_1, x_4]$ under the map $x_i \mapsto x_{2i}$ and the polynomial $f = x_3^2 - x_0x_6$. Thus the resolution is the mapping cone of the endomorphism given by multiplication with f.

$$\begin{array}{ccccccccc}
S & \longleftarrow & S^6 & \longleftarrow & S^8 & \longleftarrow & S^3 & \longleftarrow & 0\\
\uparrow{\scriptstyle f} & & \uparrow{\scriptstyle f} & & \uparrow{\scriptstyle f} & & \uparrow{\scriptstyle f} & & \\
S & \longleftarrow & S^6 & \longleftarrow & S^8 & \longleftarrow & S^3 & \longleftarrow & 0
\end{array}$$

For degree reasons, the universal unfolding has a 2-valued section and the flat family will have an exact subcomplex of type

$$S \longleftarrow S^5 \longleftarrow S^8 \longleftarrow S^5 \longleftarrow S^1 \longleftarrow 0.$$

Jan Stevens (personal communication) showed that L', like $L = \{6, 9, 13, 16\}$, defines a smoothable curve singularity that is not smoothable in a positively

graded family, and is thus not Weierstrass; indeed the 2-valued section defines a pair of double points in the general positively graded family. Similar constructions yield a lot of Weierstrass semigroups whose families have a multivalued section.

The previous example is closely related to Torres' construction [Tor94] of non-Weierstrass semigroups, of which the following is a straightforward extension. For the reader's convenience we give a proof.

Theorem 4.2. *Let L be a non-Weierstrass semigroup and let d be a prime. Any semigroup L' generated by dL and a set X of sufficiently large integers such that d does not divide every $x \in X$ is a non-Weierstrass semigroup.*

Proof. Suppose L' is the Weierstrass semigroup of a point p in a smooth projective curve $\widetilde{C}$. Let n be the maximal element among the minimal generators of L. We assume that the new generators of L' are larger than dn. Consider the morphism $\varphi_{|dnp|}\colon \widetilde{C} \to \mathbb{P}^r$ where $r = |\{\ell \in L \mid \ell < n\}|$.

If $\varphi_{|dnp|}$ were birational onto its image then the genus of $\widetilde{C}$ would be limited by Castelnuovo's bound for the genus of non-degenerate curves of degree dn in $\mathbb{P}^r$ [EH24, Theorem 10.4]. But if the integers in X are sufficiently large, then the genus of the semigroup L' exceeds this bound. Thus the morphism $\varphi_{|dnp|}$ must factor through a morphism ψ to a smooth projective curve C. Since $d = \gcd(dL)$ is prime, the degree of ψ would be d, and p would be a point of total ramification. Thus $\psi(p) \in C$ would have L as its Weierstrass semigroup, a contradiction. □

Example 4.3. For a specific instance of Theorem 4.2 we may take the semigroup L to be $\{6, 9, 13, 16\}$, which has genus 13 and, with $d = 2$, we take $L' = \{12, 18, 26, 32, 2x + 1\}$ with $x > 55$. By [EH24, Theorem 10.4], the genus of a non-degenerate curve of degree $dn = 32$ in $\mathbb{P}^r$ is bounded by

$$\pi_0(dn, r) = (r-1)M(M-1)/2 + M\epsilon$$

where $M = \lfloor (dn-1)/(r-1) \rfloor$ and $\epsilon = (dn-1) - (r-1)M$. Taking $r = 6, d = 2, n = 16$ we have $\pi_0(dn, r) = 81$, but counting the gaps we see that the genus of L' is $g' = 2 \cdot 13 + x > 81$ for $x > 55$, proving that L' is not Weierstrass.

This is not optimal: Taking $dn = 38$ it can be improved to $g' > 80$ for $x > 54$.

Remark 4.4. (1) There exist non-Weierstrass semigroups of multiplicity m for every $m \geq 6$. Examples generalizing Buchweitz's construction provide non-Weierstrass semigroups of multiplicity m and genus $m + 3$ for every $m \geq 13$ [Kom98, Section 2]. The degree-special semigroups of multiplicity $m \in \{6, \ldots, 12\}$ of smallest genus are shown in Table 3.

Multiplicity	Generators	Genus
6	$\{6, 9, 13, 16\}$	13
7	$\{7, 15, 16, 24\}$	22
8	$\{8, 9, 12, 13\}$	13
9	$\{9, 10, 14, 15\}$	17
10	$\{10, 11, 15, 16\}$	21
11	$\{11, 12, 17, 18\}$	26
12	$\{12, 13, 15, 29\}$	27
12	$\{12, 14, 15, 25\}$	27

TABLE 3. Degree-special semigroups of multiplicity $m \in \{6, \dots, 12\}$ of smallest genus, with non-Weierstrass semigroups of multiplicity $m \geq 13$ supplied by Buchweitz's construction.

(2) Kunz and Waldi [KW17] studied a family of semigroups S such that when S is minimally generated by 4 elements the resolution of the semigroup algebra has format $\{1, 6, 8, 3\}$. Moreover the ideal of a 4-generator Kunz-Waldi semigroup can be written as a sum of 3 perfect, 3-generator ideals [SS24, Sect. 4.1]. The special semigroup with generators $\{6, 9, 13, 16\}$ is Kunz-Waldi, but the special semigroup $\{6, 9, 14, 17\}$, for example, is not.

5. SEMIGROUPS OF GENUS < 13

To show that all 1413 semigroups of genus < 13 are Weierstrass, we used Pinkham's Theorem [Pin74] and computed explicit one-parameter positively graded smoothing families over $\mathbb{Q}$ directions for each of the 632 examples that was not already known. This shows that they are Weierstrass semigroups over $\mathbb{Q}$, and thus over $\mathbb{C}$.

We exhibit these families in the auxiliary file to the arXiv preprint of this paper or in the GitHub repository **eisenbud/WeierstrassSemigroups** https://github.com/eisenbud/WeierstrassSemigroups. In these places we also provide a *Macaulay2* program **allSemigroupsOfGenusLessThan13AreWeierstrass.m2** that checks flatness and smoothness for each of them. To run the code takes about 10 minutes.

Here is an outline of the computation we made to find a smoothing family for each semigroup. We begin the constructions over a finite field $\mathbb{Z}/p$ for computational efficiency. To aid in the eventual lifting to characteristic 0 we took p on the order of 10^7.

(1) We form the graded universal unfolding of the semigroup ideal, adding only positive degree parameters.

An alternate approach is to use Nathan Ilten's *Macaulay2* package VersalDeformations [Ilt, Ilt12]

Restricting to deformation parameters in a certain range of degrees, for example degrees $> b$, can make the computation easier, and we were able to do this in most cases.

(2) We use Gröbner bases to compute equations on the parameters that guarantee the constancy of the Gröbner basis in the family, and thus flatness. Once such a family is found we prune the family by eliminating the parameters that are polynomial functions of other parameters, c.f. [Hau83].
(3) Decompose the base of the flat family into irreducible components and prune the family over each component again.
(4) Pick a random point in each component and check the smoothness of the fiber over that point.
(5) Checking smoothness of a fiber: If the set of generators L of the semigroup has n elements, then the affine curve $\widetilde{C}$ lies in $\mathbb{A}^n$ and may have large codimension. Hence computing all the minors of order $(n-1)$ of the Jacobian matrix M of the ideal $I(\widetilde{C})$ is often infeasible. We exploit the fact that the fibers are 1-dimensional as follows:
    (a) We find an ideal J generated by some of the minors of order $n-1$ of M that does not vanish identically on $\widetilde{C}$. For this, we need to find minors F not vanishing identically on $\widetilde{C}$. To do this we look at the easier problem of finding complete intersections $n-1$ homogeneous equations in the ideal of the semigroup. The minors of the corresponding equations of $\widetilde{C}$ often give us the minors we need.
    (b) At this point, $\widetilde{C}$ is smooth outside the reduced points where J vanishes. Represent the radical of $J+I(\widetilde{C})$ as an intersection of maximal ideals corresponding to points on $\widetilde{C}$. It now suffices to compute the rank of the Jacobian matrix at each of the residue fields at these points, and computing the rank of a matrix over a field is easy.
(6) Once we have found a flat smoothing family in characteristic p over an irreducible component B of the base, we try to lift it to a flat one-parameter family defined over $\mathbb{Q}$.
    (a) We try to find a point in B with coordinates representable by integers of small absolute value.
        (i) If the ideal of B is 0, we choose a point with small coefficients arbitrarily.

(ii) We look for a subspace defined by $\dim B$ variables whose common zero locus is a subspace V of B.
(iii) If the corresponding projection map

$$\pi_V : B \smallsetminus V \to \mathbb{A}^{\dim B}$$

is birational, then we can find a point $b \in B$ as the preimage of a point with small coordinates in $\mathbb{A}^{\dim B}$.
(iv) We then consider the fiber C over b.

(b) (Clearing denominators) We multiply each equation of C by a small integer and check whether, after multiplying, the coefficients can be represented by integers of moderate size. If so we move these equations to a polynomial ring over $\mathbb{Q}$ and homogenize with an additional variable z.
(c) We then check whether the resulting family over $\operatorname{Spec}\mathbb{Q}[z]$ is flat.
(d) If this is the case check whether the fiber over $z = 1$ is a smooth curve.

The bottlenecks in this computation are computing the flattening relations of a (restricted) unfolding, pruning its ideal, decomposing the base into components; and checking smoothness.

Remark 5.1. Step (6.a) is successful amazingly often. In fact for every one of the 632 semigroups of genus < 13 that were not previously known to be Weierstrass with $B \neq 0$, we were able to find a subspace V, possibly after varying the base component B, and the choice of b.

A possible reason for this success is that the base of the deformation space is defined by a homogeneous ideal in a polynomial ring where the variables have many different degrees. Quite often the largest degree variables occur only once and in quadratic terms of the equations, so that their values are determined by the values of the lower degree variables. The fact that they often occur in quadratic monomials might come from the fact that the obstruction map is quadratic.

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